MAPS FROM A FORMAL SPECTRUM INTO AN AFFINE SCHEME ARE THE SAME AS MAPS FROM THE SPECTRUM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof span cites one same-chapter lemma, delivered as a navigation stub. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Morphisms of formal spectra correspond to continuous ring maps with the target given the discrete topology. Same chapter, Lemma lemma-morphism-between-formal-spectra, invoked at proof lines 7194-7197. Turns morphisms Spf(A) -> Spec(B) into continuous S-algebra maps B -> A.
- The weakly admissible topological S-algebra whose formal spectrum is used. Same chapter, Definition definition-weakly-admissible. Fixes the object Spf(A) and the topology on A.

Source reference. Original lemma 0AN0: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2355–2355. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let A be a weakly admissible topological S-algebra. Let X be an affine scheme over S. Then the natural map*

$$\mathrm{Mor}_S(\mathrm{Spec}(A), X) \longrightarrow \mathrm{Mor}_S(\mathit{Spf}(A), X)$$

is bijective.

Proof. If X is affine, say $X = \mathrm{Spec}(B)$, then we see from Lemma 0AN0 that morphisms $\mathrm{Spf}(A) \to \mathrm{Spec}(B)$ correspond to continuous S-algebra maps $B \to A$ where B has the discrete topology. These are just S-algebra maps, which correspond to morphisms $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$. □

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